

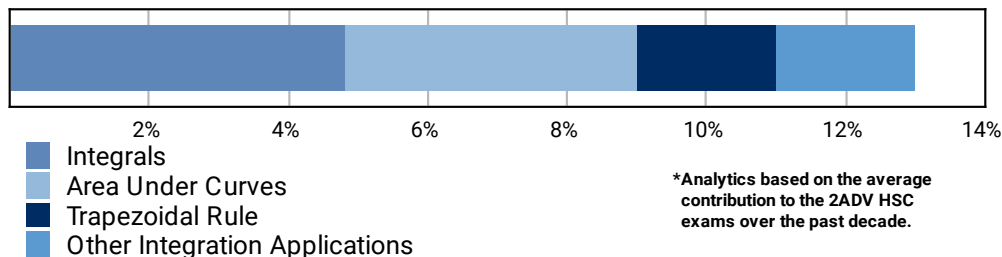
## ADV: Calculus (Adv), C4 Integration (Adv)

### Areas Under Curves (Y12)

Teacher: Denis Vlismas

Exam Equivalent Time: 157.5 minutes (based on HSC allocation of 1.5 minutes approx. per mark)

### C4 Integration



### HISTORICAL CONTRIBUTION

- C4 Integration has contributed a substantial 13.0% to past Advanced exams, on average over the past decade.
- This topic has been split into four sub-topics for analysis purposes: 1-Integrals (4.8%), 2-Areas Under Curves (4.2%), 3-Trapezoidal Rule (2.0%) and 4-Other Integration Examples (2.0%).
- This analysis looks at Areas Under Curves.

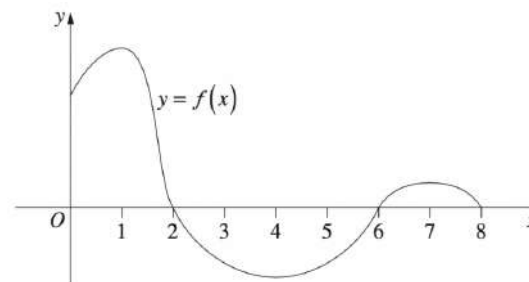
### HSC ANALYSIS - What to expect and common pitfalls

- Areas Under Curves (4.2%) are a hugely important sub-topic within the Advanced course and easily warrants significant revision attention.
- History suggests that students can count on *at least* one question appearing in the 2021 exam (with a better than even chance of two questions).
- This topic area can be examined using a very broad range of underlying equations, including quadratic, cubic, exponential, hyperbola and trig functions.
- The most common question type easily receives the highest mark allocation and requires students to find areas between equations and lines (although the area between 2 quadratics was asked in 2020 and 2017).
- 2021 combined Area Under Curves with transformations in a novel question that caused a majority of students problems. Review 2021 Q28 carefully!
- Definite integrals defining areas for functions that cannot be integrated has been asked via multiple choice 3 times in the last 6 years - all producing sub-50% mean marks. Revision attention required.

### Questions

#### 1. Calculus, 2ADV C4 2012 HSC 10 MC

The graph of  $y = f(x)$  has been drawn to scale for  $0 \leq x \leq 8$ .

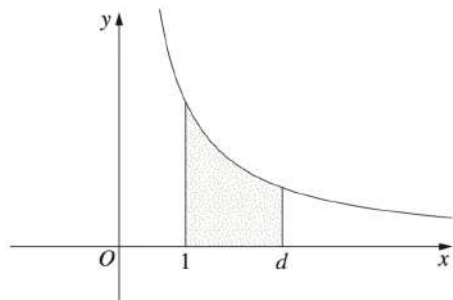


Which of the following integrals has the greatest value?

- (A)  $\int_0^1 f(x) dx$
- (B)  $\int_0^2 f(x) dx$
- (C)  $\int_0^7 f(x) dx$
- (D)  $\int_0^8 f(x) dx$

2. Calculus, 2ADV C4 2015 HSC 10 MC

The diagram shows the area under the curve  $y = \frac{2}{x}$  from  $x = 1$  to  $x = d$ .

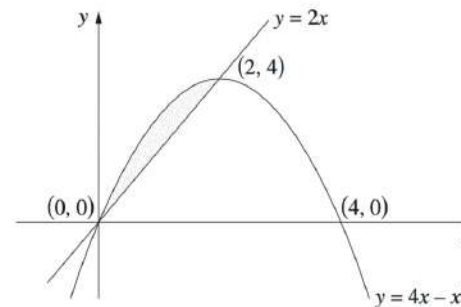


What value of  $d$  makes the shaded area equal to **2**?

- (A)  $e$
  - (B)  $e + 1$
  - (C)  $2e$
  - (D)  $e^2$
- 

3. Calculus, 2ADV C4 2015 HSC 7 MC

The diagram shows the parabola  $y = 4x - x^2$  meeting the line  $y = 2x$  at  $(0, 0)$  and  $(2, 4)$ .

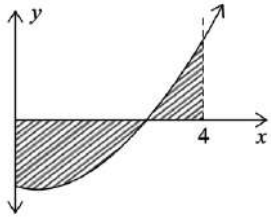


Which expression gives the area of the shaded region bounded by the parabola and the line?

- (A)  $\int_0^2 x^2 - 2x \, dx$
  - (B)  $\int_0^2 2x - x^2 \, dx$
  - (C)  $\int_0^4 x^2 - 2x \, dx$
  - (D)  $\int_0^4 2x - x^2 \, dx$
-

4. Calculus, 2ADV C4 SM-Bank 1 MC

The diagram below is the graph of  $y = x^2 - x - 6$

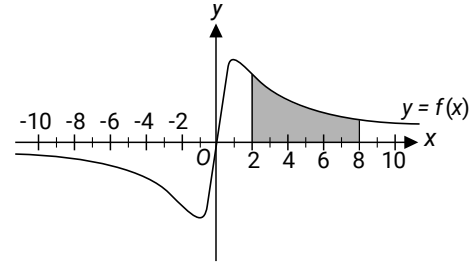


What is the correct expression for the area bounded by the  $x$ -axis and the graph  $y = x^2 - x - 6$  between  $0 \leq x \leq 4$ ?

- (A)  $A = \int_0^4 x^2 - x - 6 \, dx$
- (B)  $A = \left| \int_0^3 x^2 - x - 6 \, dx \right| + \int_3^4 x^2 - x - 6 \, dx$
- (C)  $A = \int_0^3 x^2 - x - 6 \, dx + \left| \int_3^4 x^2 - x - 6 \, dx \right|$
- (D)  $A = \left| \int_0^4 x^2 - x - 6 \, dx \right|$

5. Calculus, 2ADV C4 2019 MET-N 15 MC

The area bounded by the graph of  $y = f(x)$ , the line  $x = 2$ , the line  $x = 8$  and the  $x$ -axis, as shaded in the diagram below, is  $3 \log_e(13)$

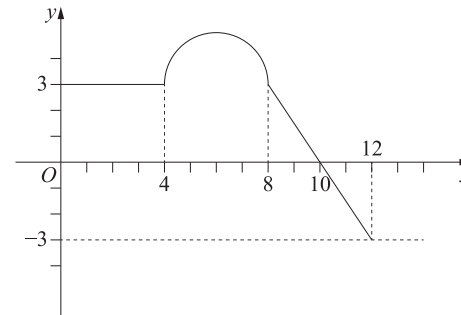


The value of  $\int_4^{10} 3f(x-2) \, dx$  is

- A.  $3 \log_e(13)$
- B.  $9 \log_e(13)$
- C.  $6 \log_e(39)$
- D.  $9 \log_e(11)$

6. Calculus, 2ADV C4 2020 HSC 7 MC

The diagram show the graph  $y = f(x)$ , which is made up of line segments and a semicircle.

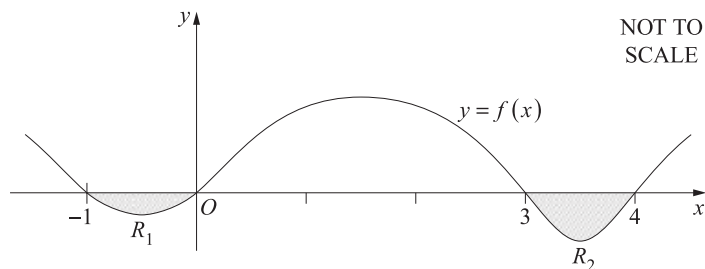


What is the value of  $\int_0^{12} f(x) \, dx$ ?

- A.  $24 + 2\pi$
- B.  $24 + 4\pi$
- C.  $30 + 2\pi$
- D.  $30 + 4\pi$

7. Calculus, 2ADV C4 2018 HSC 7 MC

The diagram shows the graph of  $y = f(x)$  with intercepts at  $x = -1, 0, 3$  and  $4$ .



The area of shaded region  $R_1$  is 2.

The area of shaded region  $R_2$  is 3.

It is given that  $\int_0^4 f(x) dx = 10$ .

What is the value of  $\int_{-1}^3 f(x) dx$ ?

- (A) 5
- (B) 9
- (C) 11
- (D) 15

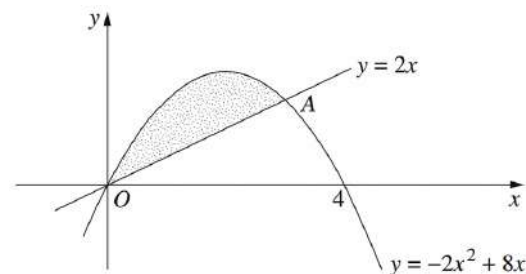
8. Calculus, 2ADV C4 2016 HSC 9 MC

What is the value of  $\int_{-3}^2 |x + 1| dx$ ?

- (A)  $\frac{5}{2}$
- (B)  $\frac{11}{2}$
- (C)  $\frac{13}{2}$
- (D)  $\frac{17}{2}$

9. Calculus, 2ADV C4 2014 HSC 12d

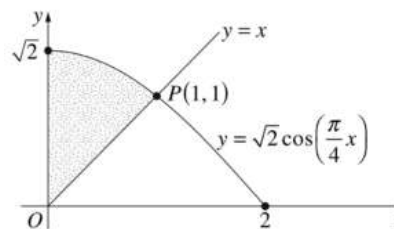
The parabola  $y = -2x^2 + 8x$  and the line  $y = 2x$  intersect at the origin and at the point  $A$ .



- i. Find the  $x$ -coordinate of the point  $A$ . (1 mark)
- ii. Calculate the area enclosed by the parabola and the line. (3 marks)

10. Calculus, 2ADV C4 2016 HSC 13d

The curve  $y = \sqrt{2} \cos\left(\frac{\pi}{4}x\right)$  meets the line  $y = x$  at  $P(1, 1)$ , as shown in the diagram.

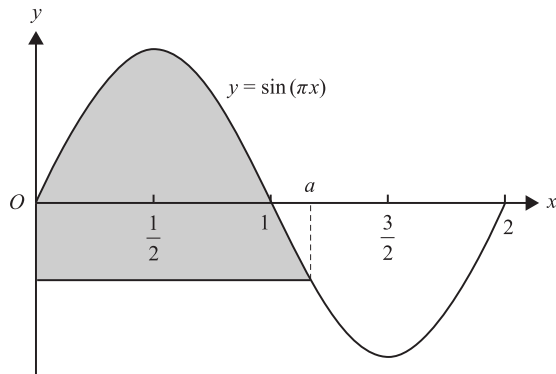


Find the exact value of the shaded area. (3 marks)

### 11. Calculus, 2ADV C4 2019 MET1 7

The shaded region in the diagram below is bounded by the vertical axis, the graph of the function with rule  $f(x) = \sin(\pi x)$  and the horizontal line segment that meets the graph at  $x = a$ , where

$$1 \leq a \leq \frac{3}{2}.$$



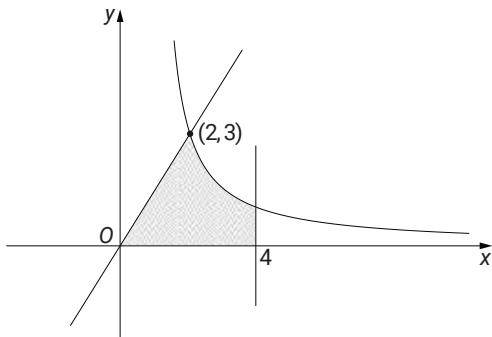
Let  $A(a)$  be the area of the shaded region.

Show that  $A(a) = \frac{1}{\pi} - \frac{1}{\pi} \cos(a\pi) - a \sin(a\pi)$ . (3 marks)

### 12. Calculus, 2ADV C4 2021 HSC 24

The curve  $y = \frac{3}{x-1}$  intersects the line  $y = \frac{3}{2}x$  at the point (2, 3).

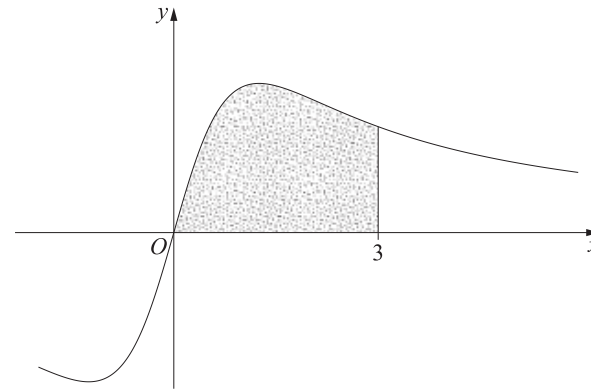
The region bounded by the curve  $y = \frac{3}{x-1}$ , the line  $y = \frac{3}{2}x$ , the  $x$ -axis and the line  $x = 4$  is shaded in the diagram.



Find the exact area of the shaded region. (3 marks)

### 13. Calculus, 2ADV C4 2019 HSC 12d

The diagram shows the graph of  $y = \frac{3x}{x^2 + 1}$ .

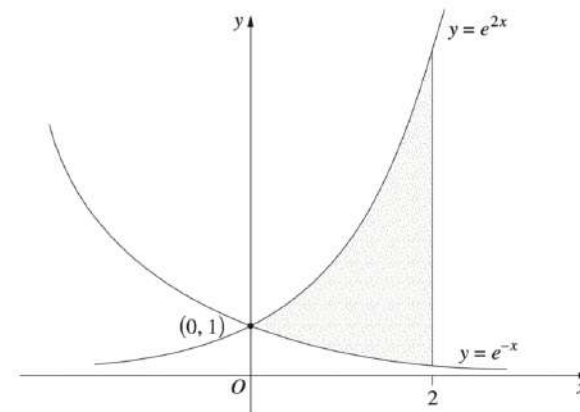


The region enclosed by the graph, the  $x$ -axis and the line  $x = 3$  is shaded.

Calculate the exact value of the area of the shaded region. (3 marks)

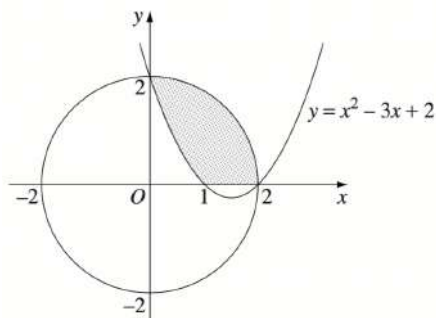
### 14. Calculus, 2ADV C4 2010 HSC 4b

The curves  $y = e^{2x}$  and  $y = e^{-x}$  intersect at the point (0, 1) as shown in the diagram.



Find the exact area enclosed by the curves and the line  $x = 2$ . (3 marks)

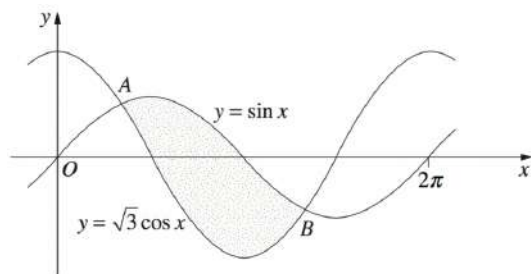
15. Calculus, 2ADV C4 2005 HSC 8b



The shaded region in the diagram is bounded by the circle of radius 2 centred at the origin, the parabola  $y = x^2 - 3x + 2$ , and the  $x$ -axis.

By considering the difference of two areas, find the area of the shaded region. (3 marks)

16. Calculus, 2ADV C4 2007 HSC 7b

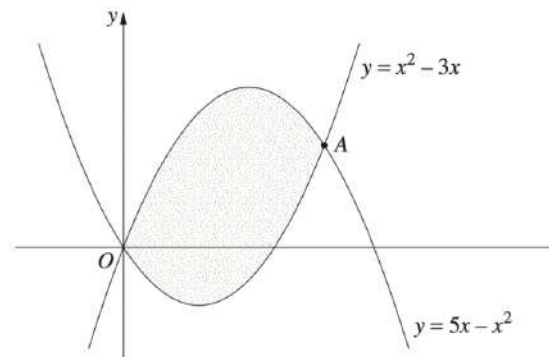


The diagram shows the graphs of  $y = \sqrt{3} \cos x$  and  $y = \sin x$ . The first two points of intersection to the right of the  $y$ -axis are labelled  $A$  and  $B$ .

- Solve the equation  $\sqrt{3} \cos x = \sin x$  to find the  $x$ -coordinates of  $A$  and  $B$ . (2 marks)
- Find the area of the shaded region in the diagram. (3 marks)

17. Calculus, 2ADV C4 2012 HSC 13b

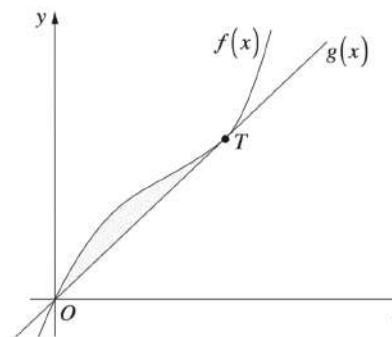
The diagram shows the parabolas  $y = 5x - x^2$  and  $y = x^2 - 3x$ . The parabolas intersect at the origin  $O$  and the point  $A$ . The region between the two parabolas is shaded.



- Find the  $x$ -coordinate of the point  $A$ . (1 mark)
- Find the area of the shaded region. (3 marks)

18. Calculus, 2ADV C4 2013 HSC 13b

The diagram shows the graphs of the functions  $f(x) = 4x^3 - 4x^2 + 3x$  and  $g(x) = 2x$ . The graphs meet at  $O$  and at  $T$ .

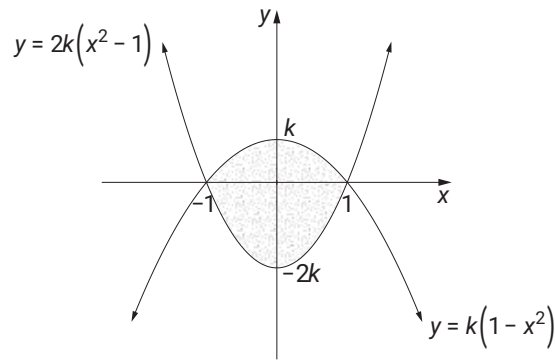


- Find the  $x$ -coordinate of  $T$ . (1 mark)
- Find the area of the shaded region between the graphs of the functions  $f(x)$  and  $g(x)$ . (3 marks)

### 19. Calculus, 2ADV C4 2017 HSC 14d

The shaded region shown is enclosed by two parabolas, each with  $x$ -intercepts at  $x = -1$  and  $x = 1$ .

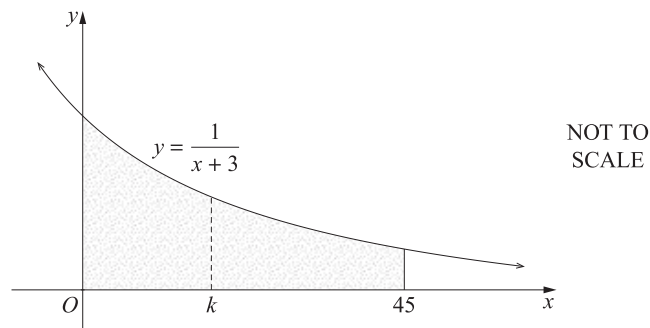
The parabolas have equations  $y = 2k(x^2 - 1)$  and  $y = k(1 - x^2)$ , where  $k > 0$ .



Given that the area of the shaded region is 8, find the value of  $k$ . (3 marks)

### 20. Calculus, 2ADV C4 2018 HSC 15b

The diagram shows the region bounded by the curve  $y = \frac{1}{x+3}$  and the lines  $x = 0$ ,  $x = 45$  and  $y = 0$ . The region is divided into two parts of equal area by the line  $x = k$ , where  $k$  is a positive integer.



What is the value of the integer  $k$ , given that the two parts have equal areas? (3 marks)

### 21. Trigonometry, 2ADV T3 2006 HSC 7b

A function  $f(x)$  is defined by  $f(x) = 1 + 2 \cos x$ .

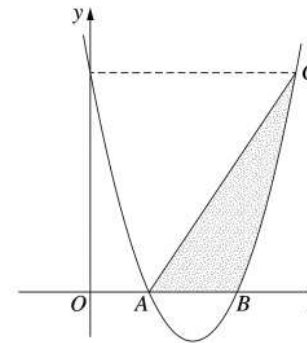
i. Show that the graph of  $y = f(x)$  cuts the  $x$ -axis at  $x = \frac{2\pi}{3}$ . (1 mark)

ii. Sketch the graph of  $y = f(x)$  for  $-\pi \leq x \leq \pi$  showing where the graph cuts each of the axes. (3 marks)

iii. Find the area under the curve  $y = f(x)$  between  $x = -\frac{\pi}{2}$  and  $x = \frac{2\pi}{3}$ . (3 marks)

### 22. Calculus, 2ADV C4 2015 HSC 16a

The diagram shows the curve with equation  $y = x^2 - 7x + 10$ . The curve intersects the  $x$ -axis at points **A** and **B**. The point **C** on the curve has the same  $y$ -coordinate as the  $y$ -intercept of the curve.



i. Find the  $x$ -coordinates of points **A** and **B**. (1 mark)

ii. Write down the coordinates of **C**. (1 mark)

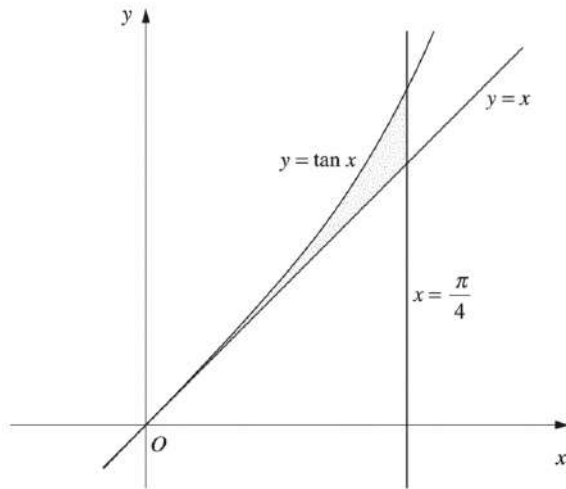
iii. Evaluate  $\int_0^2 (x^2 - 7x + 10) dx$ . (1 mark)

iv. Hence, or otherwise, find the area of the shaded region. (2 marks)

23. Calculus, 2ADV C4 2006 HSC 5b

i. Show that  $\frac{d}{dx} \log_e(\cos x) = -\tan x$ . (1 mark)

ii.

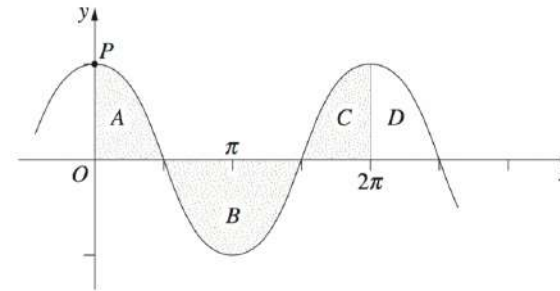


The shaded region in the diagram is bounded by the curve  $y = \tan x$  and the lines  $y = x$  and  $x = \frac{\pi}{4}$ .

Using the result of part (i), or otherwise, find the area of the shaded region. (3 marks)

24. Calculus, 2ADV C4 2011 HSC 6c

The diagram shows the graph  $y = 2 \cos x$ .



i. State the coordinates of  $P$ . (1 mark)

ii. Evaluate the integral  $\int_0^{\frac{\pi}{2}} 2 \cos x \, dx$ . (2 marks)

iii. Indicate which area in the diagram,  $A$ ,  $B$ ,  $C$  or  $D$ , is represented by the integral

$$\int_{\frac{3\pi}{2}}^{2\pi} 2 \cos x \, dx. \quad (1 \text{ mark})$$

iv. Using parts (ii) and (iii), or otherwise, find the area of the region bounded by the curve  $y = 2 \cos x$  and the  $x$ -axis, between  $x = 0$  and  $x = 2\pi$ . (1 mark)

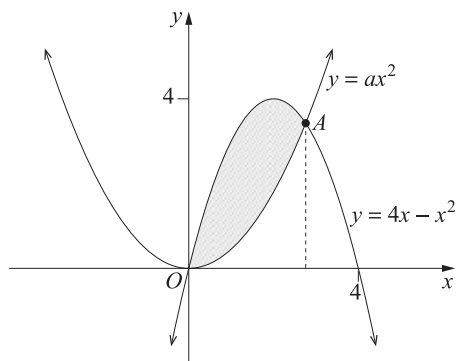
v. Using the parts above, write down the value of

$$\int_{\frac{\pi}{2}}^{2\pi} 2 \cos x \, dx. \quad (1 \text{ mark})$$



25. Calculus, 2ADV C4 2020 HSC 30

The diagram shows two parabolas  $y = 4x - x^2$  and  $y = ax^2$ , where  $a > 0$ . The two parabolas intersect at the origin,  $O$ , and at  $A$ .

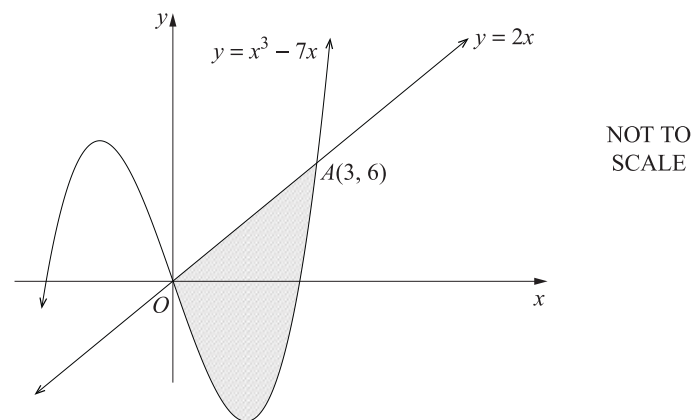


a. Show that the  $x$ -coordinate of  $A$  is  $\frac{4}{a+1}$ . (2 marks)

b. Find the value of  $a$  such that the shaded area is  $\frac{16}{3}$ . (4 marks)

26. Calculus, 2ADV C4 2018\* HSC 15c

The shaded region is enclosed by the curve  $y = x^3 - 7x$  and the line  $y = 2x$ , as shown in the diagram. The line  $y = 2x$  meets the curve  $y = x^3 - 7x$  at  $O(0, 0)$  and  $A(3, 6)$ . Do NOT prove this.



i. Use integration to find the area of the shaded region. (2 marks)

ii. Use the Trapezoidal rule and four function values to approximate the area of the shaded region. (2 marks)

The point  $P$  is chosen on the curve  $y = x^3 - 7x$  so that the tangent at  $P$  is parallel to the line  $y = 2x$  and the  $x$ -coordinate of  $P$  is positive

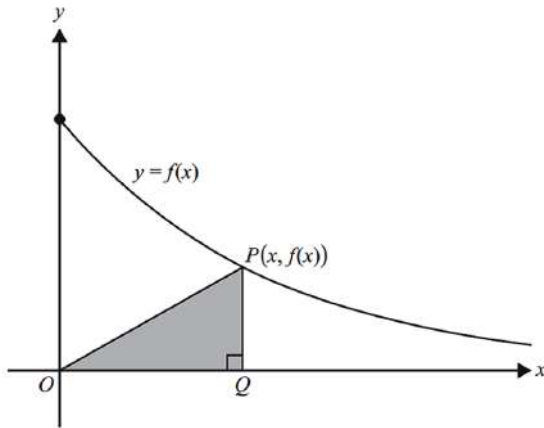
iii. Show that the coordinates of  $P$  are  $(\sqrt{3}, -4\sqrt{3})$ . (2 marks)

iv. Using the perpendicular distance formula  $\frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$ , find the area of  $\triangle OAP$ . (2 marks)

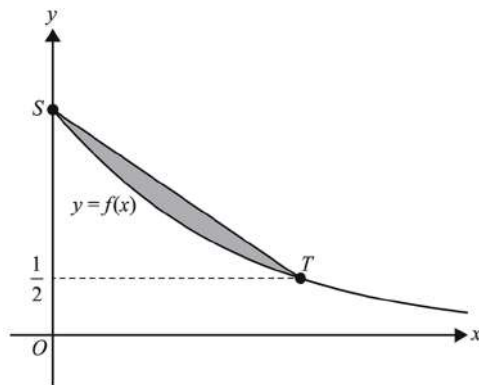
27. Calculus, 2ADV C4 SM-Bank 12

Let  $f(x) = 2e^{-\frac{x}{5}}$  for  $x \geq 0$

A right-angled triangle  $OQP$  has vertex  $O$  at the origin, vertex  $Q$  on the  $x$ -axis and vertex  $P$  on the graph of  $f$ , as shown. The coordinates of  $P$  are  $(x, f(x))$ .



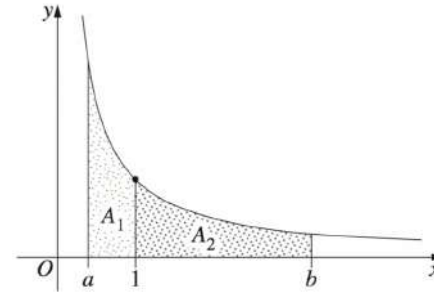
- Find the area,  $A$ , of the triangle  $OPQ$  in terms of  $x$ . (1 mark)
- Find the maximum area of triangle  $OQP$  and the value of  $x$  for which the maximum occurs. (3 marks)
- Let  $S$  be the point on the graph of  $f$  on the  $y$ -axis and let  $T$  be the point on the graph of  $f$  with the  $y$ -coordinate  $\frac{1}{2}$ . Find the area of the region bounded by the graph of  $f$  and the line segment  $ST$ . (2 marks)



28. Calculus, 2ADV C4 2010 HSC 5c

The diagram shows the curve  $y = \frac{1}{x}$ , for  $x > 0$ .

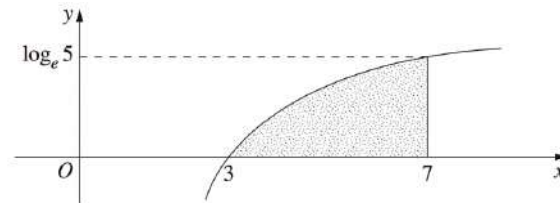
The area under the curve between  $x = a$  and  $x = 1$  is  $A_1$ . The area under the curve between  $x = 1$  and  $x = b$  is  $A_2$ .



The areas  $A_1$  and  $A_2$  are each equal to 1 square unit.

Find the values of  $a$  and  $b$ . (3 marks)

29. Calculus, 2ADV C4 2008 HSC 10a

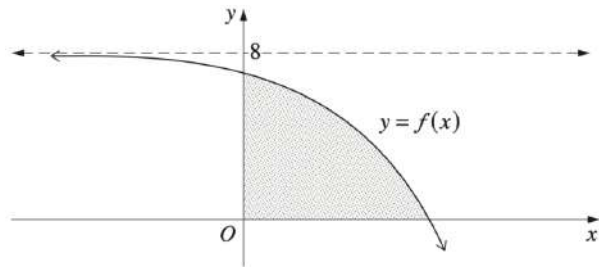


In the diagram, the shaded region is bounded by  $y = \log_e(x-2)$ , the  $x$ -axis and the line  $x = 7$ .

Find the exact value of the area of the shaded region. (5 marks)

### 30. Calculus, 2ADV C4 2021 HSC 28

The region bounded by the graph of the function  $f(x) = 8 - 2^x$  and the coordinate axes is shown



- a. Show that the exact area of the shaded region is given by  $24 - \frac{7}{\ln 2}$ . (3 marks)
- b. A new function  $g(x)$  is found by taking the graph of  $y = -f(-x)$  and translating it by 5 units to the right.  
Sketch the graph of  $y = g(x)$  showing the  $x$ -intercept and the asymptote. (2 marks)
- c. Hence, find the exact value of  $\int_2^5 g(x) dx$ . (1 mark)

### Worked Solutions

#### 1. Calculus, 2ADV C4 2012 HSC 10 MC

Since the integrals measure the net area under the graph and above the  $x$ -axis (i.e. below the  $x$ -axis is a negative value.)

$\Rightarrow B$

#### 2. Calculus, 2ADV C4 2015 HSC 10 MC

$$\int_1^d \frac{2}{x} dx = 2$$

$$[2 \log_e x]_1^d = 2$$

$$2 \log_e d - 2 \log_e 1 = 2$$

$$2 \log_e d = 2$$

$$\log_e d = 1$$

$$\therefore d = e$$

$\Rightarrow A$

#### 3. Calculus, 2ADV C4 2015 HSC 7 MC

$$\text{Shaded Area} = \int_0^2 4x - x^2 dx - \int_0^2 2x dx$$

$$= \int_0^2 4x - x^2 - 2x dx$$

$$= \int_0^2 2x - x^2 dx$$

$\Rightarrow B$

4. Calculus, 2ADV C4 SM-Bank 1 MC

$$y = x^2 - x - 6$$

$$= (x - 3)(x + 2)$$

Graph intersects the  $x$ -axis at  $(3, 0)$

$$\therefore A = \left| \int_0^3 x^2 - x - 6 \, dx \right| + \int_3^4 x^2 - x - 6 \, dx$$

$$\Rightarrow B$$

5. Calculus, 2ADV C4 2019 MET-N 15 MC

$$A_1 = \int_2^8 f(x) \, dx = 3 \log_e 13$$

$f(x - 2) = f(x)$  shifted 2 units to right.

$$\therefore \int_2^8 f(x) \, dx = \int_4^{10} f(x - 2) \, dx$$

$$\therefore 3 \int_4^{10} f(x - 2) \, dx = 3 \times 3 \log_e 13$$

$$= 9 \log_e 13$$

$\Rightarrow B$

6. Calculus, 2ADV C4 2020 HSC 7 MC

Consider the interval between  $x = 8$  and  $x = 12$ :

Area above and below the  $x$ -axis are equal.

$$\int_8^{12} f(x) \, dx = 0$$

♦ Mean mark 48%.

$$\therefore \int_0^{12} f(x) \, dx = \int_0^8 f(x) \, dx$$

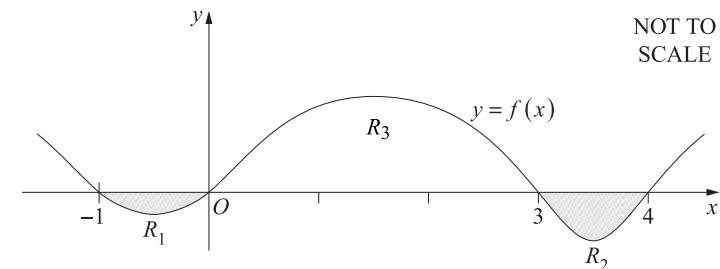
$$= \text{Area of rectangle} + \text{area of semi-circle}$$

$$= 8 \times 3 + \frac{1}{2} \times \pi \times 2^2$$

$$= 24 + 2\pi$$

$\Rightarrow A$

7. Calculus, 2ADV C4 2018 HSC 7 MC



$$\int_0^4 f(x) \, dx = 10$$

$$\therefore R_3 - R_2 = 10$$

$$R_3 = 13$$

♦ Mean mark 36%.

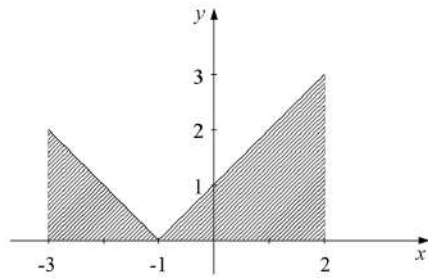
$$\int_{-1}^3 f(x) \, dx = R_3 - R_1$$

$$= 13 - 2$$

$$= 11$$

$\Rightarrow C$

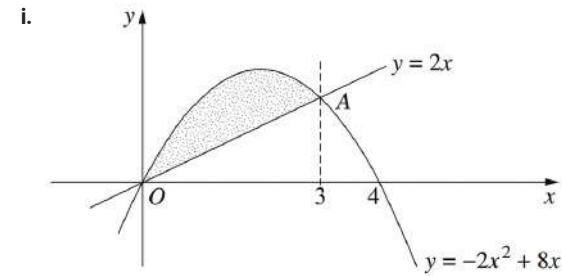
8. Calculus, 2ADV C4 2016 HSC 9 MC



♦♦♦ Mean mark 19%.

$$\begin{aligned}\int_{-3}^2 |x + 1| dx &= \text{Area of 2 triangles} \\ &= \frac{1}{2} \times 2 \times 2 + \frac{1}{2} \times 3 \times 3 \\ &= \frac{13}{2} \\ &\Rightarrow C\end{aligned}$$

9. Calculus, 2ADV C4 2014 HSC 12d



Need to find  $x$ -co-ord of  $A$

$$y = 2x \quad \dots (i)$$

$$y = -2x^2 + 8x \quad \dots (ii)$$

Subst  $y = 2x$  from (i) into (ii)

$$-2x^2 + 8x = 2x$$

$$-2x^2 + 6x = 0$$

$$-2x(x - 3) = 0$$

$$\therefore x = 0 \text{ or } 3$$

$\therefore x$ -coordinate of  $A$  is 3

ii.

$$\begin{aligned}\text{Area} &= \int_0^3 (-2x^2 + 8x) dx - \int_0^3 2x dx \\ &= \int_0^3 (-2x^2 + 6x) dx \\ &= \left[ -\frac{2}{3}x^3 + 3x^2 \right]_0^3 \\ &= \left[ \left( -\frac{2}{3}(3^3) + 3(3^2) \right) - (0 + 0) \right] \\ &= -18 + 27 \\ &= 9\end{aligned}$$

10. Calculus, 2ADV C4 2016 HSC 13d

Shaded Area

$$\begin{aligned}
 &= \int_0^1 \sqrt{2} \cos\left(\frac{\pi}{4}x\right) dx - \int_0^1 x dx \\
 &= \int_0^1 \left(\sqrt{2} \cos\left(\frac{\pi}{4}x\right) - x\right) dx \\
 &= \left[\sqrt{2} \times \frac{4}{\pi} \sin\left(\frac{\pi}{4}x\right) - \frac{x^2}{2}\right]_0^1 \\
 &= \left[\left(\frac{4\sqrt{2}}{\pi} \sin \frac{\pi}{4} - \frac{1}{2}\right) - 0\right] \\
 &= \frac{4\sqrt{2}}{\pi} \times \frac{1}{\sqrt{2}} - \frac{1}{2} \\
 &= \left(\frac{4}{\pi} - \frac{1}{2}\right) \text{ u}^2
 \end{aligned}$$

11. Calculus, 2ADV C4 2019 MET1 7

Lower border of shaded area:  $y = f(a) = \sin(a\pi)$

$$\begin{aligned}
 \text{Area} &= \int_0^a \sin(\pi x) - \sin(a\pi) dx \\
 &= \left[-\frac{1}{\pi} \cos(\pi x) - x \sin(a\pi)\right]_0^a \\
 &= \left[-\frac{1}{\pi} \cos(\pi a) - a \sin(a\pi) - \left(-\frac{1}{\pi} - 0\right)\right] \\
 &= \frac{1}{\pi} - \frac{1}{\pi} \cos(a\pi) - a \sin(a\pi)
 \end{aligned}$$

12. Calculus, 2ADV C4 2021 HSC 24

$$\begin{aligned}
 \text{Area} &= \int_0^2 \frac{3}{2}x dx + \int_2^4 \frac{3}{x-1} dx \\
 &= \left[\frac{3}{4}x^2\right]_0^2 + 3[\log_e(x-1)]_2^4 \\
 &= \frac{12}{4} + 3[\log_e 3 - \log_e 1] \\
 &= 3 + 3 \log_e 3 \text{ u}^2
 \end{aligned}$$

13. Calculus, 2ADV C4 2019 HSC 12d

$$\begin{aligned}
 \text{Area} &= \int_0^3 \frac{3x}{x^2+1} dx \\
 &= \frac{3}{2} \int_0^3 \frac{2x}{x^2+1} dx \\
 &= \frac{3}{2} [\ln(x^2+1)]_0^3 \\
 &= \frac{3}{2} (\ln 10 - \ln 1) \\
 &= \frac{3}{2} \ln 10 \text{ u}^2
 \end{aligned}$$

14. Calculus, 2ADV C4 2010 HSC 4b

$$\begin{aligned}
 \text{Area} &= \int_0^2 e^{2x} dx - \int_0^2 e^{-x} dx \\
 &= \int_0^2 (e^{2x} - e^{-x}) dx \\
 &= \left[\frac{1}{2}e^{2x} + e^{-x}\right]_0^2 \\
 &= \left[\left(\frac{1}{2}e^4 + e^{-2}\right) - \left(\frac{1}{2}e^0 + e^0\right)\right] \\
 &= \frac{1}{2}e^4 + e^{-2} - \frac{3}{2} \text{ u}^2
 \end{aligned}$$

**MARKER'S COMMENT:** The best responses used only a *single integral* before any substitution as shown in Worked Solutions.

15. Calculus, 2ADV C4 2005 HSC 8b

Shaded Area = Area in the quarter circle less  
the area below the parabola between  $x = 0$  and 1.

$$\begin{aligned}\text{Area of } \frac{1}{4} \text{ circle} &= \frac{1}{4} \times \pi r^2 \\ &= \frac{1}{4} \times \pi \times 2^2 \\ &= \pi \text{ u}^2\end{aligned}$$

Area below the parabola between  $x = 0$  and 1

$$\begin{aligned}&= \int_0^1 y \, dx \\ &= \int_0^1 x^2 - 3x + 2 \, dx \\ &= \left[ \frac{x^3}{3} - \frac{3}{2}x^2 + 2x \right]_0^1 \\ &= \left[ \left( \frac{1}{3} - \frac{3}{2} + 2 \right) - 0 \right] \\ &= \frac{5}{6}\end{aligned}$$

$$\therefore \text{Shaded Area} = \left( \pi - \frac{5}{6} \right) \text{ u}^2$$

16. Calculus, 2ADV C4 2007 HSC 7b

i.  $\sqrt{3} \cos x = \sin x$

$$\tan x = \sqrt{3}$$

Since  $\tan \frac{\pi}{3} = \sqrt{3}$  and  $\tan$  is

positive in 1st/3rd quadrants:

$$\begin{aligned}x &= \frac{\pi}{3}, \pi + \frac{\pi}{3} \\ &= \frac{\pi}{3}, \frac{4\pi}{3}\end{aligned}$$

$\therefore$  The  $x$ -coordinates of  $A$  and  $B$

$$\text{are } x = \frac{\pi}{3}, \frac{4\pi}{3}.$$

ii. Shaded Area

$$\begin{aligned}&= \int_{\frac{\pi}{3}}^{\frac{4\pi}{3}} \sin x \, dx - \int_{\frac{\pi}{3}}^{\frac{4\pi}{3}} \sqrt{3} \cos x \, dx \\ &= \int_{\frac{\pi}{3}}^{\frac{4\pi}{3}} \sin x - \sqrt{3} \cos x \, dx \\ &= \left[ -\cos x - \sqrt{3} \sin x \right]_{\frac{\pi}{3}}^{\frac{4\pi}{3}} \\ &= \left[ \left( -\cos \frac{4\pi}{3} - \sqrt{3} \sin \frac{4\pi}{3} \right) - \left( -\cos \frac{\pi}{3} - \sqrt{3} \sin \frac{\pi}{3} \right) \right] \\ &= \left[ \left( \frac{1}{2} + \sqrt{3} \times \frac{\sqrt{3}}{2} \right) - \left( -\frac{1}{2} - \sqrt{3} \times \frac{\sqrt{3}}{2} \right) \right] \\ &= [2 - (-2)] \\ &= 4 \text{ u}^2\end{aligned}$$

## 17. Calculus, 2ADV C4 2012 HSC 13b

i.  $y = x^2 - 3x \dots (1)$

$y = 5x - x^2 \dots (2)$

Solve:  $(1) = (2)$

$$x^2 - 3x = 5x - x^2$$

$$2x^2 - 8x = 0$$

$$2x(x - 4) = 0$$

$$x = 0 \text{ or } 4$$

$\therefore$   $x$ -coordinate of  $A$  is 4

ii.  $\text{Area} = \int_0^4 (5x - x^2) dx - \int_0^4 (x^2 - 3x) dx$

$$= \int_0^4 (5x - x^2 - x^2 + 3x) dx$$

$$= \int_0^4 (8x - 2x^2) dx$$

$$= \left[ 4x^2 - \frac{2}{3}x^3 \right]_0^4$$

$$= \left[ (4 \times 4^2) - \left( \frac{2}{3} \times 4^3 \right) \right]$$

$$= \left[ 64 - \frac{128}{3} \right]$$

$$= \frac{64}{3} \text{ u}^2$$

**MARKER'S COMMENT:** Less errors were made in part (ii) by students who simplified the definite integral before integrating, as shown in the worked solution.

## 18. Calculus, 2ADV C4 2013 HSC 13b

i.  $T$  occurs when  $f(x) = g(x)$

$$4x^3 - 4x^2 + 3x = 2x$$

$$4x^3 - 4x^2 + x = 0$$

$$x(4x^2 - 4x + 1) = 0$$

$$x(2x - 1)^2 = 0$$

$$2x - 1 = 0 \Rightarrow x = \frac{1}{2}$$

Substitute  $x = \frac{1}{2}$  into  $g(x)$

$$g\left(\frac{1}{2}\right) = 2 \times \frac{1}{2} = 1$$

$$\therefore T\left(\frac{1}{2}, 1\right)$$

ii.  $\text{Shaded Area} = \int_0^{\frac{1}{2}} (f(x) - g(x)) dx$

$$= \int_0^{\frac{1}{2}} (4x^3 - 4x^2 + x) dx$$

$$= \left[ x^4 - \frac{4}{3}x^3 + \frac{1}{2}x^2 \right]_0^{\frac{1}{2}}$$

$$= \left[ \left( \left( \frac{1}{2} \right)^4 - \frac{4}{3} \left( \frac{1}{2} \right)^3 + \frac{1}{2} \left( \frac{1}{2} \right)^2 \right) - 0 \right]$$

$$= \frac{1}{16} - \frac{1}{6} + \frac{1}{8}$$

$$= \frac{1}{48} \text{ u}^2$$

$\therefore$  Shaded area between the curves is  $\frac{1}{48} \text{ u}^2$



19. Calculus, 2ADV C4 2017 HSC 14d

$$\text{Area} = \int_{-1}^1 k(1 - x^2) dx - \int_{-1}^1 2k(x^2 - 1) dx$$

$$8 = 2 \int_0^1 k - kx^2 - 2kx^2 + 2k dx$$

$$8 = 2 \int_0^1 3k - 3kx^2 dx$$

$$8 = 2[3kx - kx^3]_0^1$$

$$8 = 2[(3k - k) - 0]$$

$$8 = 4k$$

$$\therefore k = 2$$

20. Calculus, 2ADV C4 2018 HSC 15b

$$\begin{aligned} \text{Total Area} &= \int_0^{45} \frac{1}{x+3} \\ &= [\ln(x+3)]_0^{45} \\ &= \ln 48 - \ln 3 \\ &= \ln 16 \end{aligned}$$

$$\Rightarrow \int_0^k \frac{1}{x+3} = \frac{1}{2} \times \ln 16$$

$$[\ln(x+3)]_0^k = \ln 16^{\frac{1}{2}}$$

$$\ln(k+3) - \ln 3 = \ln 4$$

$$\ln\left(\frac{k+3}{3}\right) = \ln 4$$

$$\frac{k+3}{3} = 4$$

$$\begin{aligned} \therefore k &= 4 \times 3 - 3 \\ &= 9 \end{aligned}$$

21. Trigonometry, 2ADV T3 2006 HSC 7b

i.  $f(x) = 1 + 2 \cos x$

$f(x)$  cuts the  $x$ -axis when  $f(x) = 0$

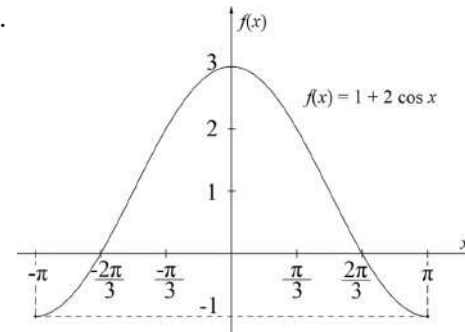
$$1 + 2 \cos x = 0$$

$$2 \cos x = -1$$

$$\cos x = -\frac{1}{2}$$

$$\therefore x = \frac{2\pi}{3} \dots \text{as required}$$

ii.



iii.  $\text{Area} = \int_{-\frac{\pi}{2}}^{\frac{2\pi}{3}} 1 + 2 \cos x dx$

$$= [x + 2 \sin x]_{-\frac{\pi}{2}}^{\frac{2\pi}{3}}$$

$$= \left[ \left( \frac{2\pi}{3} + 2 \sin \frac{2\pi}{3} \right) - \left( -\frac{\pi}{2} + 2 \sin -\frac{\pi}{2} \right) \right]$$

$$= \left( \frac{2\pi}{3} + 2 \times \frac{\sqrt{3}}{2} \right) - \left( -\frac{\pi}{2} + 2(-1) \right)$$

$$= \frac{2\pi}{3} + \sqrt{3} + \frac{\pi}{2} + 2$$

$$= \left( \frac{7\pi}{6} + \sqrt{3} + 2 \right) \text{ u}^2$$

i.  $y = x^2 - 7x + 10$   
 $= (x - 2)(x - 5)$

$$\therefore x = 2 \text{ or } 5$$

$$\therefore x\text{-coordinate of } A = 2$$

$$x\text{-coordinate of } B = 5$$

ii.  $y$  intercept occurs when  $x = 0$

$$\Rightarrow y\text{-intercept} = 10$$

$C$  occurs at intercept:

$$y = x^2 - 7x + 10 \quad \dots (1)$$

$$y = 10 \quad \dots (2)$$

$$(1) = (2)$$

$$x^2 - 7x + 10 = 10$$

$$x^2 - 7x = 10$$

$$x(x - 7) = 10$$

$$x = 0 \text{ or } 7$$

$$\therefore C \text{ is } (7, 10)$$

iii.  $\int_0^2 (x^2 - 7x + 10) dx$

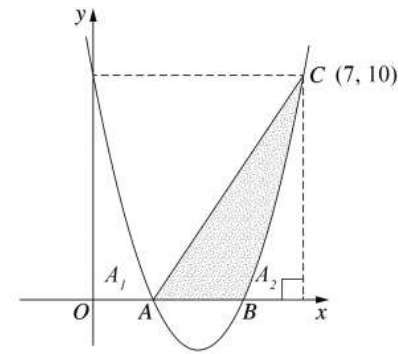
$$= \left[ \frac{1}{3}x^3 - \frac{7}{2}x^2 + 10x \right]_0^2$$

$$= \left[ \left( \frac{1}{3} \times 2^3 - \frac{7}{2} \times 2^2 + 10 \times 2 \right) - 0 \right]$$

$$= \frac{8}{3} - 14 + 20$$

$$= 8\frac{2}{3}$$

iv.



$$A_1 = A_2$$

$$A_2 = 8\frac{2}{3} \text{ u}^2 \quad (\text{from part (iii)})$$

Let  $D$  be  $(7, 0)$

Shaded Area

$$= \text{Area of } \triangle ACD - A_2$$

$$= \frac{1}{2}bh - 8\frac{2}{3}$$

$$= \frac{1}{2} \times 5 \times 10 - 8\frac{2}{3}$$

$$= 16\frac{1}{3} \text{ u}^2$$

♦ Mean mark 49%.

23. Calculus, 2ADV C4 2006 HSC 5b

$$\begin{aligned} \text{i. } \frac{d}{dx} \log_e(\cos x) &= \frac{-\sin x}{\cos x} \\ &= -\tan x \dots \text{as required} \end{aligned}$$

ii. Shaded Area

$$\begin{aligned} &= \int_0^{\frac{\pi}{4}} \tan x \, dx - \int_0^{\frac{\pi}{4}} x \, dx \\ &= \int_0^{\frac{\pi}{4}} \tan x - x \, dx \\ &= \left[ -\log_e(\cos x) - \frac{1}{2}x^2 \right]_0^{\frac{\pi}{4}} \\ &= \left[ \left( -\log_e\left(\cos\frac{\pi}{4}\right) - \frac{1}{2} \times \frac{\pi^2}{16} \right) - \left( -\log_e(\cos 0) - 0 \right) \right] \\ &= -\log_e \frac{1}{\sqrt{2}} - \frac{\pi^2}{32} + \log_e 1 \\ &= -\log_e 2^{-\frac{1}{2}} - \frac{\pi^2}{32} \\ &= \left( \frac{1}{2} \log_e 2 - \frac{\pi^2}{32} \right) \text{u}^2 \end{aligned}$$

24. Calculus, 2ADV C4 2011 HSC 6c

$$\begin{aligned} \text{i. } y &= 2 \cos x \\ \text{At } x &= 0 \\ y &= 2 \cos 0 = 2 \\ \therefore P(0, 2) \end{aligned}$$

$$\begin{aligned} \text{ii. } \int_0^{\frac{\pi}{2}} 2 \cos x \, dx \\ &= [2 \sin x]_0^{\frac{\pi}{2}} \\ &= \left[ 2 \sin\left(\frac{\pi}{2}\right) - 2 \sin 0 \right] \\ &= 2 - 0 \\ &= 2 \end{aligned}$$

iii.  $C$

$$\begin{aligned} \text{iv. Since Area } A &= \text{Area } C, \text{ and} \\ \text{Area } B &= 2 \times \text{Area } A \end{aligned}$$

$$\begin{aligned} \therefore \text{Total Area} &= 2 + (2 \times 2) + 2 \\ &= 8 \text{ u}^2 \end{aligned}$$

$$\begin{aligned} \text{v. } \int_{\frac{\pi}{2}}^{2\pi} 2 \cos x \, dx \\ &= \text{Area } C - \text{Area } B \\ &= 2 - (2 \times 2) \\ &= -2 \end{aligned}$$

**MARKER'S COMMENT:** "Using the parts above" in part (v) was ignored by many students. Important to know when finding an area and evaluating an integral may differ.

a. Intersection occurs when

$$4x - x^2 = ax^2$$

$$x^2(a+1) - 4x = 0$$

$$x[x(a+1) - 4] = 0$$

$$x(a+1) - 4 = 0 \quad \text{or} \quad x = 0$$

$$\therefore x_A = \frac{4}{a+1}$$

♦ Mean mark part (b) 48%.

$$\text{b. Area} = \int_0^{\frac{4}{a+1}} 4x - x^2 \, dx - \int_0^{\frac{4}{a+1}} ax^2 \, dx$$

$$\frac{16}{3} = \int_0^{\frac{4}{a+1}} 4x - (1+a)x^2 \, dx$$

$$\frac{16}{3} = \left[ 2x^2 - \left( \frac{1+a}{3} \right) x^3 \right]_0^{\frac{4}{a+1}}$$

$$\frac{16}{3} = 2 \left( \frac{4}{a+1} \right)^2 - \left( \frac{1+a}{3} \right) \left( \frac{4}{a+1} \right)^3$$

$$\frac{16}{3} = \frac{32}{(a+1)^2} - \frac{64}{3} \cdot \frac{1}{(a+1)^2}$$

$$16 = \frac{96}{(a+1)^2} - \frac{64}{(a+1)^2}$$

$$16(a+1)^2 = 32$$

$$(a+1)^2 = 2$$

$$a+1 = \sqrt{2}, \quad (a > 0)$$

$$\therefore a = \sqrt{2} - 1$$

$$\text{i. Area} = \int_0^3 2x - (x^3 - 7x) \, dx$$

$$= \int_0^3 9x - x^3 \, dx$$

$$= \left[ \frac{9}{2}x^2 - \frac{1}{4}x^4 \right]_0^3$$

$$= \left[ \left( \frac{9}{2} \times 3^2 - \frac{1}{4} \times 3^4 \right) - 0 \right]$$

$$= \frac{81}{2} - \frac{81}{4}$$

$$= \frac{81}{4} \text{ units}^2$$

$$\text{ii. } f(x) = 9x - x^3$$

|        |   |   |    |   |
|--------|---|---|----|---|
| $x$    | 0 | 1 | 2  | 3 |
| $f(x)$ | 0 | 8 | 10 | 0 |
| weight | 1 | 2 | 2  | 1 |

$$\text{Area} \approx \frac{1}{2} [0 + 2(8 + 10) + 0]$$

$$\approx \frac{1}{2} (36)$$

$$\approx 18 \text{ u}^2$$

$$\text{iii. } y = x^3 - 7x$$

$$\frac{dy}{dx} = 3x^2 - 7$$

$$\text{Find } x \text{ such that } \frac{dy}{dx} = 2$$

$$3x^2 - 7 = 2$$

$$3x^2 = 9$$

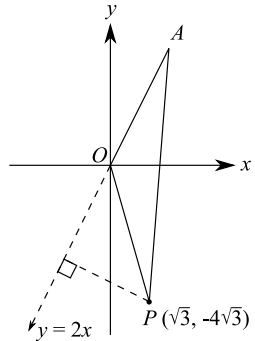
$$x^2 = 3$$

$$x = \sqrt{3} \quad (x > 0)$$

$$\begin{aligned}
 y &= (\sqrt{3})^3 - 7\sqrt{3} \\
 &= 3\sqrt{3} - 7\sqrt{3} \\
 &= -4\sqrt{3}
 \end{aligned}$$

$\therefore P$  has coordinates  $(\sqrt{3}, -4\sqrt{3})$

iv.



$$\begin{aligned}
 \text{dist } OA &= \sqrt{(3-0)^2 + (6-0)^2} \\
 &= \sqrt{45} \\
 &= 3\sqrt{5}
 \end{aligned}$$

Find  $\perp$  distance of  $P$  from  $OA$

$$P(\sqrt{3}, -4\sqrt{3}), \quad 2x - y = 0$$

$$\begin{aligned}
 \perp \text{ dist} &= \left| \frac{2\sqrt{3} + 4\sqrt{3}}{\sqrt{3} + 2} \right| \\
 &= \frac{6\sqrt{3}}{\sqrt{5}}
 \end{aligned}$$

$$\begin{aligned}
 \therefore \text{Area} &= \frac{1}{2} \times 3\sqrt{5} \times \frac{6\sqrt{3}}{\sqrt{5}} \\
 &= 9\sqrt{3} \text{ units}^2
 \end{aligned}$$

27. Calculus, 2ADV C4 SM-Bank 12

$$\begin{aligned}
 \text{i. Area} &= \frac{1}{2} \times b \times h \\
 &= \frac{1}{2} x \left( 2e^{-\frac{x}{5}} \right) \\
 \therefore A &= xe^{-\frac{x}{5}}
 \end{aligned}$$

ii. Stationary point when  $\frac{dA}{dx} = 0$ ,

$$\begin{aligned}
 x \left( -\frac{1}{5} e^{-\frac{x}{5}} \right) + e^{-\frac{x}{5}} &= 0 \\
 e^{-\frac{x}{5}} \left( 1 - \frac{x}{5} \right) &= 0 \\
 \therefore x &= 5 \quad \left( e^{-\frac{x}{5}} > 0, \text{ for all } x \right)
 \end{aligned}$$

$$\text{When } x = 5, \quad A = xe^{-\frac{x}{5}} = 5e^{-1}$$

$$\therefore A_{\max} = \frac{5}{e} \text{ u}^2, \text{ when } x = 5$$

iii. Find  $S$ :  $F(0) = 2$

$$\Rightarrow S(0, 2)$$

$$\text{Find } T: \quad 2e^{-\frac{x}{5}} = \frac{1}{2}$$

$$e^{-\frac{x}{5}} = \frac{1}{4}$$

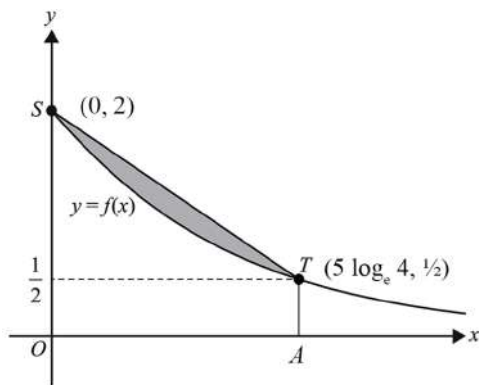
$$-\frac{x}{5} = \log_e \left( \frac{1}{4} \right)$$

$$x = 5 \log_e(4)$$

$$\Rightarrow T \left( 5 \log_e(4), \frac{1}{2} \right)$$

♦ Mean mark (Vic) 35%.

♦♦ Mean mark (Vic) 32%.



$$\begin{aligned}
 \therefore \text{Area} &= \text{Area } SOAT - \int_0^{5 \log_e(4)} \left( 2e^{-\frac{x}{5}} \right) dx \\
 &= \frac{1}{2}h(a+b) + 10 \left[ e^{-\frac{x}{5}} \right]_0^{5 \log_e(4)} \\
 &= \frac{5}{2} \log_e(4) \left( 2 + \frac{1}{2} \right) + 10 \left[ e^{-\log_e(4)} - e^0 \right] \\
 &= \frac{25}{4} \log_e(4) + 10 \left( \frac{1}{4} - 1 \right) \\
 &= \frac{25}{4} \log_e(4) - \frac{15}{2} \text{ u}^2
 \end{aligned}$$

28. Calculus, 2ADV C4 2010 HSC 5c

$$\int_a^1 \frac{1}{x} dx = 1$$

$$[\ln x]_a^1 = 1$$

$$\ln 1 - \ln a = 1$$

$$\ln a = -1$$

$$\therefore a = e^{-1} = \frac{1}{e}$$

$$\int_1^b \frac{1}{x} dx = 1$$

$$[\ln x]_1^b = 1$$

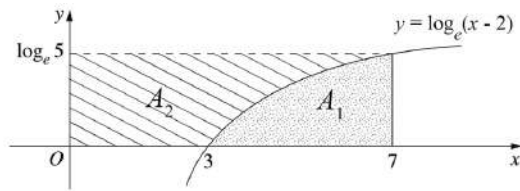
$$\ln b - \ln 1 = 1$$

$$\ln b = 1$$

$$\therefore b = e$$

**IMPORTANT:** Note when  $\log_e a = 1$ , the definition of a log means that  $e^1 = a$ . Many students failed to earn an easy 3rd mark by recognising this.

29. Calculus, 2ADV C4 2008 HSC 10a



Shaded Area ( $A_1$ ) = Rectangle -  $A_2$

Area of Rectangle =  $7 \times \log_e 5$

Finding the Area of  $A_2$

$$y = \log_e(x - 2)$$

$$x - 2 = e^y$$

$$x = e^y + 2$$

$$\therefore A_2 = \int_0^{\log_e 5} x \, dy$$

$$= \int_0^{\log_e 5} e^y + 2 \, dy$$

$$= [e^y + 2y]_0^{\log_e 5}$$

$$= [(e^{\log_e 5} + 2\log_e 5) - (e^0 + 0)]$$

$$= (5 + 2\log_e 5) - 1$$

$$= 4 + 2\log_e 5$$

$$\therefore A_1 = 7\log_e 5 - (4 + 2\log_e 5)$$

$$= 5\log_e 5 - 4$$

30. Calculus, 2ADV C4 2021 HSC 28

a.  $x$ -intercept occurs when

$$8 - 2^x = 0 \Rightarrow x = 3$$

$$\text{Area} = \int_0^3 8 - 2^x \, dx$$

$$= \left[ 8x - \frac{2^x}{\ln 2} \right]_0^3$$

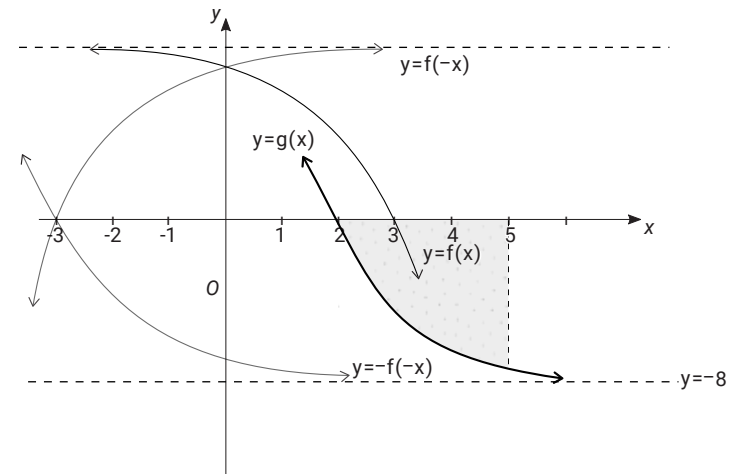
$$= 24 - \frac{8}{\ln 2} - \left( 0 - \frac{1}{\ln 2} \right)$$

$$= 24 - \frac{8}{\ln 2} + \frac{1}{\ln 2}$$

$$= 24 - \frac{7}{\ln 2}$$

b.

♦ Mean mark part (b) 48%.



$y = f(-x) \rightarrow$  reflect  $y = f(x)$  in the  $y$ -axis

$y = -f(-x) \rightarrow$  reflect  $y = f(-x)$  in the  $x$ -axis

c.  $\int_2^5 g(x) \, dx$  is the same area as found in part (a)

♦♦♦ Mean mark part (c) 13%.

except it is below the  $x$ -axis.

$$\therefore \int_2^5 g(x) \, dx = \frac{7}{\ln 2} - 24$$